Data Analysis on Market Timings vs Long-Term Investing

Final Project Report

Course: COMP 3250 – Data Analytics I
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1. Abstract

This project is a data-driven analysis project with the goal of comparing two common stock investment strategies, Buy & Hold Strategy (long term investing) and Market timing, to provide clear evidence-based support to first-time/casual investors.

Using historical ETF data retrieved from Yahoo Finance via Python, we simulated six portfolio scenarios across three distribution types (Single, Equal, and Custom) for both strategies.

The performance was evaluated using financial metrics including Compound Annual Growth Rate (CAGR), Sharpe Ratio, Volatility, Maximum Drawdown, and Total Profit.

The results are communicated through an interactive Power BI dashboard, enabling non-expert users to explore and compare strategy outcomes over a 20+ year period.

2. Project Scope & Problem description

This project addresses the following core research question:

"When it comes to investing to make a profit, are casual investors better off with market timing or long-term investing?"

Casual investors seeking to use stock investments as a source of income often face significant uncertainty when entering the stock market due. The knowledge gap between professional and casual investors leads first-time/casual investors to speculation-driven decisions, investment traps, or confusion when making decisions.

A persistent and widely debated question is whether simply investing and holding over a longer period produces better returns than attempting to time the market (buying low and selling high).

This project aims to close the knowledge gap as well as clear up speculations by comparing these two entry-level investment strategies.

The scope of this project is bounded by the following parameters:

1. Three ETFs analyzed:

- SPY: one of the largest ETFs in the U.S., designed to track the performance of the S&P 500 Index (500 of the largest publicly traded companies in the United States).

- QQQ: an ETF that tracks the Nasdaq-100 Index. It includes 100 of the largest non-financial companies listed on the Nasdaq stock exchange.

- VTI (Total Market): an ETF that provides exposure to the entire investable U.S. stock market.

2. Historical data spanning 20+ years retrieved from Yahoo Finance
3. Two strategies: Buy & Hold and Market Timing using a 50-day Moving Average signal (a technical indicator that shows the average price of a stock over the last 50 trading days).
4. Six portfolio simulations:
   - Buy & Hold Strategy using just SPY ETF,
   - Market Timing Strategy using just SPY ETF,
   - Buy & Hold Strategy using equal investments into each ETF,
   - Market Timing Strategy using equal investments into each ETF,
   - Buy & Hold Strategy using a custom investment strategy,
   - Market Timing Strategy using a custom investment strategy.
5. Initial investment of $10,000 with biweekly contributions of $500
6. Performance metrics:
   CAGR - represents the specific rate at which an investment would have grown if it had grown at a steady rate each year, with profits being reinvested at the end of each year
   Sharpe Ratio - measuring risk-adjusted return. It measures how much extra return you get for every unit of risk (volatility) you endure.
   Volatility - represents uncertainty. High volatility means prices jump around wildly in a short time; low volatility means the price is relatively stable.
   Maximum Drawdown,
   Total Profit - The raw dollar amount you gained or lost over the entire life of the investment.

3. Team Involvement and Project Organization

The project team consists of four members;each assigned a clearly defined role aligned with the Data Analytics Lifecycle (DAL) framework. Project management and task tracking were handled using Hive and communications were done on Discord.

All code and deliverables were version-controlled in a shared GitHub repository using a branching workflow.

Member	Role	Responsibilities
Ada Taing	Business/Data Analyst	Defined investment problem, selected evaluation metrics, interpreted simulation results, and ensured alignment with casual investor needs.
Adedeji Adeboyejo	Data Engineer	Retrieved ETF data via Yahoo Finance API, cleaned and prepared datasets, calculated

		indicators, and simulated all six investment strategies.
Aidan Zhu	Visualization Developer	Designed and implemented the interactive Power BI dashboard, ensuring accessibility and clarity for a non-expert audience.
Mani Manav	Documentation Lead	Maintained project documentation and prepared the final written report and video presentation.

The team followed the Data Analytics Lifecycle (DAL) as the guiding framework. The project ran from February to the last week of March 2026, organized into six weekly stages:

Period	DAL Stage	Key Tasks
Week 1	Discovery	Finalized problem statement, defined research question, set up Hive workspace, created data dictionary.
Week 2	Data Preparation	Retrieved raw ETF data, performed data cleaning and validation, built combined_prices.csv.
Week 3	Model Planning & Building	Calculated MA50 indicators, simulated all six strategy/portfolio combinations in Python.
Week 4	Model Complete	Validated simulation outputs, computed CAGR, Sharpe Ratio, Volatility, Max Drawdown, Total Profit.
Week 5	Communicate Results	Built the Power BI dashboard with interactive filters and comparative visualizations.
Final Week	Operationalize	Completed the final report and presentation and submitted all deliverables.

4. Project Design and Development

4.1 Tech Stack

• Python:
Used for all data engineering tasks including retrieval (yfinance), cleaning (pandas), indicator calculation, and strategy simulation.

• Microsoft Power BI:
Used as the primary visualization and dashboard platform, consuming CSV outputs from the Python pipeline.

• GitHub: Used for version control and collaboration; full audit trail of contributions; easy sharing of CSV outputs.

4.2 Data Sources & Data Preparation

Stage	Notebook	Source
Retrieval	data_retrieval.ipynb	Yahoo Finance API
Cleaning	data_cleaning.ipynb	data/raw/*.csv
Indicators	indicators.ipynb	combined_prices.csv
Simulations	single_buy_hold.ipynb single_timing.ipynb equal_buy_hold.ipynb equal_timing.ipynb custom_buy_hold.ipynb custom_timing.ipynb	prices_with_indicators.csv
Combine	combine_simulations.ipynb	data/simulations/*.csv

4.3 Data Preparation

The data preparation process followed a sequential pipeline implemented across three Jupyter notebooks. Each notebook has a single responsibility and passes its output to the next stage.

1. <u>Data retrieval</u>

Raw data was downloaded using yfinance with period='max' to capture all available historical data. Each ETF was saved as a separate CSV file in data/raw/.

2. <u>Cleaning</u>

The following cleaning operations were applied to each raw file:

• Date parsing and formatting: Converted Date to datetime type, then formatted as YYYY-MM-DD string for Power BI compatibility.

• Chronological sorting: Ensured rows are ordered oldest to newest before any calculations.

• Duplicate removal: Dropped any rows with duplicate dates, keeping the first occurrence.

• Missing value handling: Dropped rows with null price values; filled missing Volume with 0.

After cleaning the three individual files, they were merged into a single combined_prices.csv using an inner join on Date, so that only dates where all three ETFs have data are retained. This file contains four columns: Date, SPY_Price, QQQ_Price, and VTI_Price.

3. Data validation

The notebooks used for the project involved steps to validate that the data being loaded/processed was done directly

4. Indicators

The 50-day Moving Average (MA50) was calculated for each ETF using a rolling window of 50 trading days (.rolling(50).mean()). This produces prices_with_indicators.csv, which adds SPY_MA50, QQQ_MA50, and VTI_MA50 columns to the combined prices. The first 49 rows of each MA50 column are NaN by design — insufficient historical data exists to compute a 50-day average until row 50. These rows are preserved in the dataset and handled within the simulation logic.

5. Model Building

This step involved simulating the portfolio using jupyter notebooks

4.3 Analytical Approach

This project will investigate two different strategies which are Buy & Hold and timing the market. The analysis will determine which of the strategies will generate the most profits while taking into account the risks.

To support the comparison, the analysis will also evaluate strategy performance with these measures:

- Final Portfolio Value: the amount the portfolio has at the end of analysis
- CAGR: compunded annual growth as a percent
- Volatility: measures the level of risk as a percentage
- Sharpe Ratio: how well a portfolio performs in correspondence to every unit of risk
- Maximum Drawdown: the most the portfolio has lost

These metrics will not only analyze how much the portfolio makes but also the trade offs. The dashboard will be organized to show the comparison results visually.

4.4 Dashboard Design

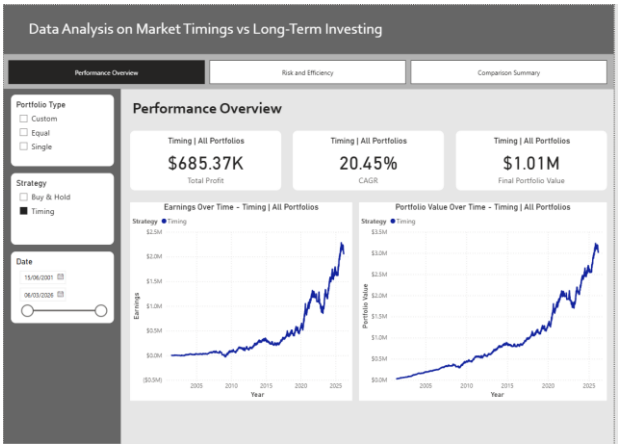
The dashboard was created using Power BI and designed to present clear results with interactive features.

Dashboard includes:

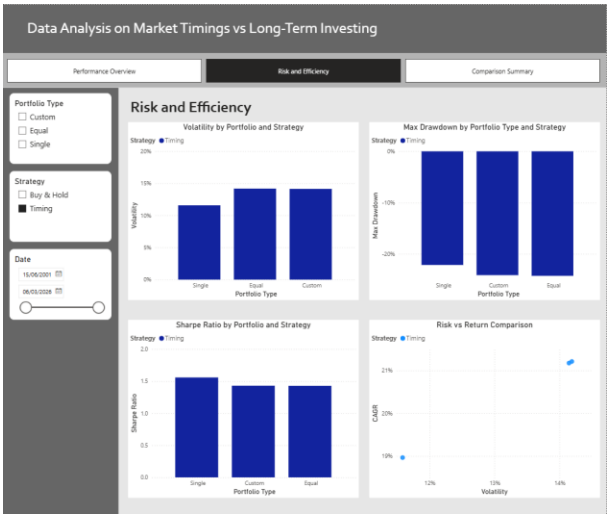
- Performance Overview
- Risk and Efficiency Overview
- Summary of comparisons

Elements in the Dashboard:

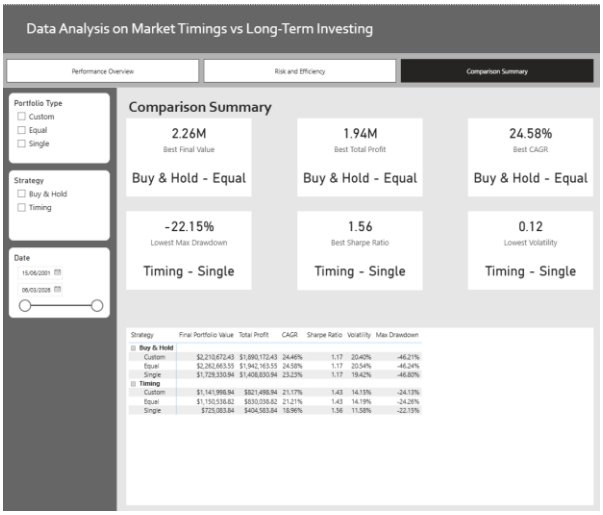
- Line charts
- Clustered Bar Charts
- Scatter plot charts
- Cards showing metric data
- Filters/slicers for portfolio type, strategy, and date
- Matrix table summarizing the performance metrics



Page 1 of the dashboard



Page 2 of the dashboard



Page 3 of the dashboard

4.5 Design Rationale

The design was chosen to effectively let the user see results clearly and quickly identify important patterns. Filters and page navigation features help the user with getting quick results by letting them focus on specific scenarios and outcomes.

Line charts were used to show time series progression easily as it allows users to easily point out high and low points within a timeline. Bar charts are also used to show a comparison in value between different portfolios and strategies with clear and simple distinctions. Cards are used as the fastest method for identifying key information as it shows one value and what the value is for.

5. Findings and Future Work

5.1 Current Insights

The completed dashboard shows that Buy & Hold creates the most value, holding the highest overall total profit, final portfolio value, and compounded growth rate. However, timing the the market is far less risky than Buy & Hold as it has the least volatility, the smallest major loss, and the more efficient Sharpe Ratio. In conclusion, Buy & Hold seems to be the better option when it comes to making money, while timing the market is the safer option if investors are afraid of losing money in large amounts.

5.2 Future Work

Some future improvements could be to explore more ETFs and investing strategies. With these additions, it could provide more insights on other strategies casual investors could use to make a profit.

6. References

Data Analytics / Visual Analytics tools:

- Power BI: https://learn.microsoft.com/en-us/power-bi/

Project Management Platform:

- Hive: https://hive.com/

Data retrieval and cleaning:

- Python: https://docs.python.org/3/tutorial/index.html

Datasets:

- Yahoo Finance:
  - https://python-yahoofinance.readthedocs.io/en/latest/api.html
  - https://ranaroussi.github.io/yfinance/
  - https://finance.yahoo.com/

7. Addendums

A: User Requirements and Analysis Document

A.1 Project Overview

This document defines the user requirements for the COMP 3250 data analysis project. The system retrieves, processes, and simulates historical ETF investment data, and communicates results through an interactive Power BI dashboard accessible to casual investors with no financial background.

A.1 Functional requirements

1. The system shall retrieve historical daily OHLCV price data for SPY, QQQ, and VTI from the Yahoo Finance API using the yfinance Python library.
2. The system shall clean raw data: standardize column names, remove duplicate dates, handle missing values, validate price logic (Low <= Close <= High), and format all dates as YYYY-MM-DD.
3. The system shall merge cleaned ETF data into a single combined_prices.csv file containing Date, SPY_Price, QQQ_Price, and VTI_Price.
4. The system shall calculate a 50-day rolling moving average for each ETF and output prices_with_indicators.csv.
5. The system shall simulate the Buy & Hold strategy for Single, Equal, and Custom portfolio distributions, starting with a $10,000 initial investment and $500 biweekly contributions.
6. The system shall simulate the Market Timing strategy for Single, Equal, and Custom portfolio distributions using the MA50 signal (buy when price > MA50, sell when price <= MA50).
7. Each simulation shall produce a CSV file with 21 columns as defined in the data dictionary, including Portfolio_Value, Earnings, Daily_Return, Strategy, and Portfolio_Type.
8. All six simulation files shall be combined into a single combined_simulations.csv for Power BI consumption.
9. The system shall calculate the following performance metrics for each simulation: CAGR, Volatility, Sharpe Ratio, Maximum Drawdown, and Total Profit.

B: Design document

B.1 Logical Architecture

• Data Layer (Python):
A sequential pipeline of Jupyter notebooks that transforms raw Yahoo Finance data into clean,

simulation-ready CSV files. Each notebook has a single responsibility and passes its output to the next.

• Presentation Layer (Power BI):
An interactive dashboard that imports the simulation CSV files and presents findings through time-series charts, comparison bar charts, metric cards, and slicers.

B.2 Physical Architecture

The project is hosted in a shared GitHub repository with the following structure:

Folder	Contents
data/raw/	Raw ETF CSV files from Yahoo Finance
data/processed/	Cleaned ETFs, combined_prices.csv, prices_with_indicators.csv
data/simulations/	Six simulation output CSVs and all_simulations.csv
notebooks/	
dashboard/	Power BI dashboard file (.pbix)
docs/	Problem Overview, Data dictionary, simulation strategy documentation, User Dashboard Guide, Simulation Approach, Performance metrics
.gitignore	Excludes venv/, .ipynb_checkpoints/, __pycache__/, *.pyc, data/raw/*.csv
requirements.txt	Python dependencies: pandas, yfinance, numpy, jupyter, openpyxl

• Documentation:

Project Report

Project Presentation

B.4 Key Data Dictionary Entries

Column	Type	Description
Date	Date (YYYY-MM-DD)	Trading session date
SPY_Price / QQQ_Price / VTI_Price	Float	Adjusted closing price for each ETF
SPY_MA50 / QQQ_MA50 / VTI_MA50	Float	50-day moving average price. NaN for first 49 rows. Blank in Buy & Hold simulations.
Position_SPY/QQQ/VTI	Integer (0 or 1)	1 = hold/buy signal. 0 = sell signal. Always 1 in Buy & Hold.
Contribution	Float	$500 every 10 rows; $0 otherwise. Does not include the initial investment.
Cash	Float	Uninvested cash. Always 0 in Buy & Hold. Non-zero in Timing when position = 0.
Shares_SPY/QQQ/VTI	Float	Shares held on that day. Only increases in Buy & Hold. Can drop to 0 in Timing.
Total_Invested	Float	Cumulative money invested into the portfolio = $10,000 + sum of all contributions to date.
Portfolio_Value	Float	Current total value of all holdings
Earnings	Float	Portfolio_Value - Total_Invested. Positive = profit; Negative = loss.
Daily_Return	Percent	Day-over-day percentage change in Portfolio_Value. NaN on first row.
Strategy	String	'Buy & Hold' or 'Timing'

Portfolio_Type	String	'Single', 'Equal', or 'Custom'

C. Deployment document

C.1 System Requirements

Component	Requirement
Operating System	Windows 10/11, macOS 12+, or Ubuntu 20.04+
Python	Version 3.9 or higher
Power BI	Microsoft Power BI Desktop (free download) Windows only. Mac users: Power BI Service via browser.
Git	Any recent version for repository access
Storage	Approximately 500 MB for all raw, processed, and simulation CSV files

C.2 Python Environment Setup

1. Clone the repository:

```
git clone https://github.com/AdaTaing/COMP3250-Project
```

2. Create and activate a virtual environment:

```
python -m venv venv
venv\Scripts\activate        # Windows
source venv/bin/activate       # Mac / Linux
```

3. Install all dependencies:

```
pip install -r requirements.txt
```

4. Launch Jupyter:

```
jupyter notebook
```

C.3 Running the Pipeline

Execute the notebooks in order from the notebooks/ folder. Each notebook must complete without errors before the next is run:

Order	Notebook	Expected Output
1	data_retrieval.ipynb	data/raw/SPY_raw.csv, QQQ_raw.csv, VTI_raw.csv
2	data_cleaning.ipynb	data/processed/SPY_cleaned.csv, QQQ_cleaned.csv, VTI_cleaned.csv, combined_prices.csv
3	indicators.ipynb	data/processed/prices_with_indicators.csv
4	single_buy_hold.ipynb single_timing.ipynb equal_buy_hold.ipynb equal_timing.ipynb custom_buy_hold.ipynb custom_timing.ipynb	data/simulations/ single_buy_hold.csv, single_timing.csv, equal_buy_hold.csv, equal_timing.csv, custom_buy_hold.csv, custom_timing.csv
5	combine_simulations.ipynb	data/simulations/combined_simulations.csv

No manual steps are required between notebooks. Each notebook reads from and writes to defined file paths that are consistent across the pipeline.

C.4 Power BI Dashboard Setup

D. User Guide

<u>D.1 Purpose</u>

This guide explains how to use the Power BI dashboard to explore and interpret the investment strategy comparison results. No financial background is required.

<u>D.2 Dashboard Overview</u>

D.3 Interpreting the Metrics

Metric	Simple Explanation	What to Look For
CAGR (%)	The average annual growth rate of the portfolio over the full period. Like a bank interest rate, but for the whole portfolio.	Higher is better. A CAGR of 10% means the portfolio grew at an average of 10% per year.
Sharpe Ratio	How much return you earned for each unit of risk taken. It rewards strategies that grow steadily rather than through big swings.	Above 1.0 is considered good. Higher means better risk-adjusted returns.
Volatility (%)	How much the portfolio value fluctuated from day to day. A volatile portfolio swings up and down more dramatically.	Lower means a smoother ride. Higher volatility means the portfolio value was less predictable.
Max Drawdown (%)	The single worst loss the portfolio experienced from its highest point to its lowest. A max drawdown of -40% means the portfolio lost 40% of its peak value at its worst moment.	Closer to 0% is better. A large max drawdown can be alarming even if the portfolio recovered.
Total Earnings ($)	The dollar profit earned above the total amount invested.	Higher is better. Negative means the portfolio lost money overall.